Unit 8: Guidelines & Heuristics for Evaluating the User Interface

Unit Learning Objectives

The unit learning outcomes are to:

1. Be able to apply guidelines and Nielsen's heuristics for User Interface Design in the design and evaluation of user interfaces

General Guidelines for UI Design

Refer to the document "UI Design Guidelines", under *Unit 7: The User Interface* on Funda, for some tips and guidelines for appropriate use and examples of HIOs.

Usability Heuristics

Heuristics are a set of general principles used in interactive design to ensure usability and provide a relatively simple and quick testing regime to identify problems and possible solutions. They are called "heuristics" because they are broad rules of thumb and not specific usability guidelines. These principles can be followed in the design and evaluation of user interfaces.

Jakob Nielsen's ten heuristics are probably the most used usability heuristics for user interface design. (https://www.nngroup.com/articles/ten-usability-heuristics/).

#1: Visibility of system status

The design should always keep users informed about what is going on, through appropriate feedback within a reasonable amount of time. When users know the current system status, they learn the outcome of their prior interactions and determine next steps. Predictable interactions create trust in the product as well as the brand.

Tips

- Communicate clearly to users what the system's state is — no action with consequences to users should be taken without informing them.
- Present feedback to the user as quickly as possible (ideally, immediately).

#2: Match between system and the real world

The design should speak the users' language. Use words, phrases, and concepts familiar to the user, rather than internal jargon. Follow real-world conventions, making information appear in a natural and logical order. The way you should design depends very much on your specific users. Terms, concepts, icons, and images that seem perfectly clear to you and your colleagues may be unfamiliar or confusing to your users.

When a design's controls follow real-world conventions and correspond to desired outcomes (called natural mapping), it's easier for users to learn and remember how the interface works. This helps to build an experience that feels intuitive.

Tips

- Ensure users can understand meaning without having to go look up a word's definition.
- Never assume your understanding of words or concepts will match those of your users.
- User research will help you uncover your users' familiar terminology, as well as their mental models around important concepts.

#3: User control and freedom

Users often perform actions by mistake. They need a clearly marked "emergency exit" to leave the unwanted action without having to go through an extended process. When it's easy for people to back out of a process or undo an action, it fosters a sense of freedom and confidence. Exits allow users to remain in control of the system and avoid getting stuck and feeling frustrated.

Tips

- Support *Undo* and *Redo*.
- Show a clear way to exit the current interaction, like a *Cancel* button.
- Make sure the exit is clearly labelled and discoverable.

#4: Consistency and standards

Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform and industry conventions. Jakob's Law states that people spend most of their time using digital products *other than yours*. Users' experiences with those other products set their expectations. Failing to maintain consistency may increase the users 'reasoning by forcing them to learn something new.

Tips

- Improve learnability by maintaining both types of consistency: internal and external.
- Maintain consistency within a single product or a family of products (internal consistency).
- Follow established industry conventions (external consistency).

#5: Error prevention

Good error messages are important, but the best designs carefully prevent problems from occurring in the first place. Either eliminate error-prone conditions, or check for them and present users with a confirmation option before they commit to the action.

There are two types of errors: slips and mistakes. Slips are unconscious errors caused by inattention. Mistakes are conscious errors based on a mismatch between the user's mental model and the design.

Tips

- Prioritize your effort: Prevent high-cost errors first, then little frustrations.
- Avoid slips by providing helpful constraints and good defaults.
- Prevent mistakes by removing memory burdens, supporting undo, and warning your users.

#6: Recognition rather than recall

Minimize the user's memory load by making elements, actions, and options visible. The user should not have to remember information from one part of the interface to another. Information required to use the design (e.g. field labels or menu items) should be visible or easily retrievable when needed.

Humans have limited short-term memories. Interfaces that promote recognition reduce the amount of cognitive effort required from users.

Tips

- Let people recognize information in the interface, rather than having to remember ("recall") it.
- Offer help in context, instead of giving users a long tutorial to memorize.
- Reduce the information that users have to remember.

#7: Flexibility and efficiency of use

Shortcuts — hidden from novice users — may speed up the interaction for the expert user such that the design can cater to both inexperienced and experienced users. Allow users to tailor frequent actions. Flexible processes can be carried out in different ways, so that people can pick whichever method works for them.

Tips

- Provide accelerators like keyboard shortcuts and touch gestures.
- Provide personalization by tailoring content and functionality for individual users.
- Allow for customization, so users can make selections about how they want the product to work.

#8: Aesthetic and minimalist design

Interfaces should not contain information which is irrelevant or rarely needed. Every extra unit of information in an interface competes with the relevant units of information and diminishes their relative visibility.

This heuristic doesn't mean you have to use a flat design — it's about making sure you're keeping the content and visual design focused on the essentials. Ensure that the visual elements of the interface support the user's primary goals.

Tips

- Keep the content and visual design of UI focus on the essentials.
- Don't let unnecessary elements distract users from the information they really need.
- Prioritize the content and features to support primary goals.

#9: Help users recognize, diagnose, and recover from errors

Error messages should be expressed in plain language (no error codes), precisely indicate the problem, and constructively suggest a solution. These error messages should also be presented with visual treatments that will help users notice and recognize them.

Tips

- Use traditional error message visuals, like bold, red text.
- Tell users what went wrong in language they will understand — avoid technical jargon.
- Offer users a solution, like a shortcut that can solve the error immediately.

#10: Help and documentation

It is best if the system doesn't need any additional explanation. However, it may be necessary to provide documentation to help users understand how to complete their tasks. Help and documentation content should be easy to search and focused on the user's task. Keep it concise, and list concrete steps that need to be carried out.

Tips

- Ensure that the help documentation is easy to search.
- Whenever possible, present the documentation in context right at the moment that the user requires it.
- List concrete steps to be carried out.

Watch the Nielsen's Heuristics videos for additional explanations on each heuristic